AI-Based BNU Vehicle Monitoring System

Sticker and Number Plate Detection using YOLOv8, OCR and SQLite

AI THEORY PROJECT

BNU GATE

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Project Snapshot

2 Detected Classes	**87** Annotated Images
89.4% Training mAP50	**95.4%** Test Precision

Prototype goal: reduce manual guard entry by automatically detecting stickers and number plates, then saving a digital log with date and time.

1. Project Context: Why This System?

Manual vehicle checking at a university gate can be improved using AI.

BNU Gate Monitoring Problem

- At the gate, a guard may manually check stickers and write vehicle details.
- During rush hours, manual checking can create delays and missed entries.
- Paper-based records are harder to search later for date/time evidence.
- An AI-based prototype can support faster verification and digital logging.

Main Aim

Build a student prototype that detects BNU sticker + number plate and stores a timestamped database record.

2. Proposed AI Solution

Object detection + OCR + database logging in one pipeline.

A. Detect the Vehicle Evidence

YOLOv8 looks at each image/frame and detects two object classes: BNU sticker and number plate.

YOLOv8

B. Read the Plate Text

After YOLO finds the plate box, the plate area is cropped and passed to OCR for text extraction.

EasyOCR

C. Save a Digital Log

SQLite stores plate number, sticker status, confidence score, date and time for each processed vehicle.

SQLite

3. Dataset and Annotation

The model learned from labeled examples of BNU stickers and number plates.

How the Dataset Was Prepared

- Images were collected from real vehicle/gate-style situations.
- Roboflow was used to draw bounding boxes around objects.
- Only two labels were used: bnu_sticker and number_plate.
- Dataset was exported in YOLOv8 format: images + .txt label files + data.yaml.

62	17	8
Train Images	Validation Images	Test Images

YOLO Label Format

Each label file stores: class_id, x_center, y_center, width, height.

Real BNU campus context used to keep the presentation visually aligned with the university use case.

4. AI Concepts and Technology Stack

The system combines theory concepts with practical implementation tools.

Computer Vision

Allows the system to process vehicle images instead of manual observation.

Object Detection

Finds exact locations of sticker and number plate using bounding boxes.

YOLOv8 Nano

Lightweight object detection model suitable for faster inference.

OCR

Reads plate text after the detected number plate area is cropped.

SQLite Database

Stores each detection record locally in a structured table.

OpenCV + Python

Handles image reading, drawing boxes, camera feed and backend logic.

Important distinction: YOLO detects the number plate area; OCR reads the text written on the plate.

5. System Workflow

End-to-end flow from image input to saved vehicle record.

Camera / Image

Input frame from gate camera or uploaded test image

YOLOv8 Detection

Detect bnu_sticker and number_plate boxes

Plate Crop

Crop only the detected plate region

OCR Reading

Extract plate text from the cropped image

SQLite Log

Save plate, sticker status, confidence and time

Final Output Record

Example: Plate Number = “BAG 842”, Sticker Detected = Yes/No, Confidence = 0.89, Date/Time = system timestamp

6. Model Training Methodology

Training was done in Google Colab using the YOLOv8 object detection pipeline.

Training Setup

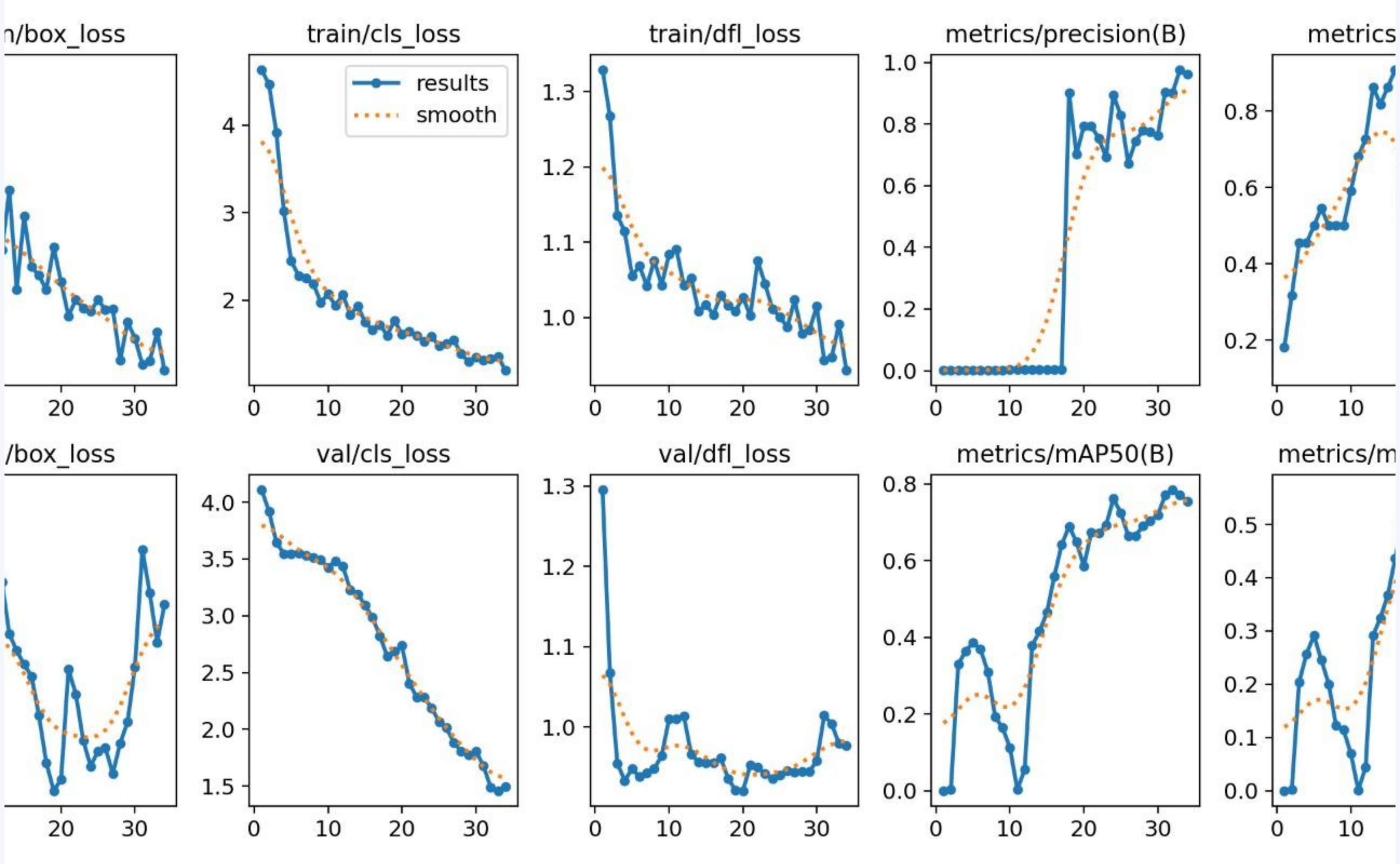
- Base model: YOLOv8n (nano) for lightweight and fast detection.
- Training platform: Google Colab with GPU support.
- Input size: 640 × 640 images for YOLO training.
- Epochs used in training run: 50 with best weights saved automatically.
- Final model file: best.pt — used later in the application for predictions.

Why YOLOv8?

It predicts bounding boxes and class labels in one forward pass, which makes it suitable for real-time vehicle monitoring.

7. Training Results Graphs

Loss decreased and precision/recall/mAP improved during training.



How to Read This Slide

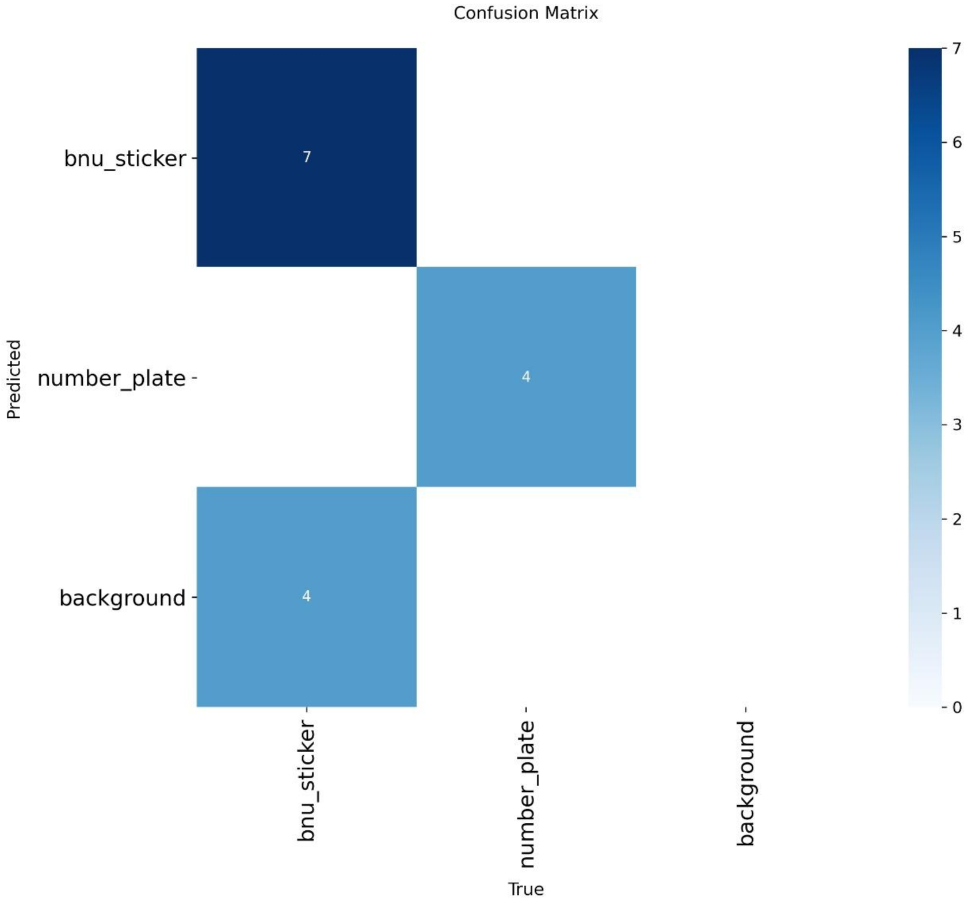
- Box loss and class loss reduce as the model learns better object boundaries and labels.
- Precision increases, meaning most predicted detections are correct.
- mAP50 improves across epochs, showing better object detection performance.
- Validation curves are not perfectly smooth because the dataset is small.

89.4%
Training / validation mAP50

95.4%
Test precision

8. Confusion Matrix and Evaluation

The confusion matrix shows correct detections and missed objects class-wise.



Observed Evaluation Points

- The model correctly detected BNU stickers and number plates in the validation/test evaluation.
- Some sticker objects were missed as background, mainly because stickers are small or unclear.
- Number plate detection was stronger than sticker detection in our experiments.
- A larger and more balanced dataset would improve recall and reduce missed detections.

Viva-safe Result Statement

This is a working prototype. Its precision is good, but real deployment needs more BNU sticker images from different angles and lighting.

9. Conclusion and Future Work

A practical AI prototype for smarter university gate monitoring.

What We Achieved

- Created a YOLOv8-based sticker and number plate detection prototype.
- Used OCR to read number plate text after detection.
- Saved vehicle logs using SQLite with date and time.
- Prepared a real workflow that can be connected to a webcam or phone camera.

Future Improvements

- Collect more BNU sticker images in sunlight, shade and night conditions.
- Improve OCR using clearer plate crops and local plate format rules.
- Connect with live gate camera for real-time demonstration.
- Add admin dashboard for search, filters and unauthorized vehicle alerts.

Thank You